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Faithfulness in the Face of Confounding: Shapley Concept Effects for LLM Explanations

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Outline

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The Correlated Concept Problem

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Original Metric (Matton et al., 2025 [2]): Causal Concept Faithfulness (CCF)

- Assumes *disentangled* concepts – no correlation.
- **In reality, concepts often co-vary!**

Two critical manifestations:

1. **Synergy:** Two concepts alone may not flip the answer, but *together* they do.
   - Example: Credit rating – "no employment" + "no property" jointly reject, but not always individually.
2. **Redundancy:** Concept C_m is removed, but the LLM still infers it from C_n.
   - Example: "Nationality" omitted, yet "Mother tongue = Hindi" $\Rightarrow$ "Indian". Explanation never mentions nationality, but decision still depends on it.

$\Rightarrow$ The original atomic CE underestimates true causal influence.

What We Need

A new faithfulness metric must:

1. **Handle correlations** – correctly attribute causal effects even when concepts interact.
2. **Backwards compatible** – produce exactly the original CCF scores when no correlations exist.
3. **Same output range** – $F(x) \in [-1, 1]$ (Pearson correlation between CE and EE).
4. **Computationally feasible** – may increase cost, but stay within practical bounds.

We call this extension **Correlation-Controlled Causal Concept Faithfulness (4CF)**.

Step 0: Assume a Correlation-Detection Oracle

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For a given question x with concept set $\mathcal{C} = \{C_0, \ldots, C_{q-1}\}$:

- An oracle partitions $\mathcal{C}$ into disjoint **correlation groups** $S_1, S_2, \ldots$ plus singleton (non-correlated) concepts.
- For any $C_m \notin \bigcup S_i$: use the original atomic CE.
- For $C_m \in S_i$: we compute a **Shapley-CE** based on the entire group S_i.

This selective approach keeps computation manageable.
(Later we will discuss how to implement the oracle.)

The Coalition Game on a Correlation Group S

Define a value function $v(T)$ for any subset $T \subseteq S$:

$$v(T) = \frac{1}{|\mathrm{alt}(T)|} \sum_{\mathrm{alt}} D_{\mathrm{KL}}\Big(P_M(Y \mid x_{T \to \mathrm{alt}}) \,\Big\|\, P_M(Y \mid x)\Big)$$

$v(\emptyset) = 0.$ It measures the joint effect of simultaneously swapping all concepts in T.

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Shapley value for $C_m \in S$ [3]:

$$\phi_m = \sum_{T \subseteq S \setminus \{m\}} \frac{|T|!\,(|S| - |T| - 1)!}{|S|!} \big[v(T \cup \{m\}) - v(T)\big]$$

This is the average marginal contribution of C_m across all possible orders of adding concepts.

Constructing the New CE Vector

For question x with q concepts:

1. Oracle partitions $\mathcal{C}$ into disjoint groups $S_1, \ldots, S_k$ and singletons.
2. For each group S, generate a $2^{|S|}$ factorial counterfactual grid and compute $v(T)$ for all $T \subseteq S$.
3. Compute Shapley values ϕ_m for each $C_m \in S$.
4. For singleton concepts, compute the original atomic CE as before.
5. Assemble the final vector

$$\mathrm{CE}_{4\mathrm{CF}}(x) = \left[\text{either } \phi_m \text{ or } \mathrm{CE}_{\mathrm{orig}}(x, C_m)\right]_{m=0}^{q-1}$$

Illustrative Example: Synergy

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- Question x about credit approval, with three concepts: **Employment** (E), **Savings** (S), **Personal Status** (P).
- Oracle: $S_1 = \{E, S\}$ are correlated and synergistic; P is independent.
- The LLM's true behavior: rejection occurs most likely when *both* E and S are bad.

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Factorial grid for S_1:

	S (orig.)	S' (swapped)
E (orig.)	(E, S)	(E, S')
E' (swapped)	(E', S)	(E', S')

Estimated v values (KL effects):

- $v(\{E\}) = 0.1$
- $v(\{S\}) = 0.2$
- $v(\{E, S\}) = 0.9$

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Shapley scores:

$$\phi_E = \tfrac{1}{2}(0.1 - 0) + \tfrac{1}{2}(0.9 - 0.2) = 0.4, \qquad \phi_S = \tfrac{1}{2}(0.2 - 0) + \tfrac{1}{2}(0.9 - 0.1) = 0.5$$

$\Rightarrow$ The synergy is fairly split; each concept receives half of the joint effect (+0.3).

Singletons and Final Vector

- For P (Personal Status): compute original atomic CE $\rightarrow$ some value $\mathrm{CE}_{\mathsf{orig}}(P)$.
- Final CE vector:

$$\mathsf{CE}_{\mathsf{4CF}} = [\,\phi_E,\ \phi_S,\ \mathrm{CE}_{\mathsf{orig}}(P)\,] = [\,0.4,\ 0.5,\ \mathrm{CE}_{\mathsf{orig}}(P)\,]$$

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Take-away: Without correlation handling, the original method would give E and S only their small independent effects (0.2 each), underestimating their true importance. 4CF correctly captures the synergy.

EE and Final Score – No Changes Needed

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- **Explanation-Implied Effect (EE):** still defined as the probability that the explanation mentions the concept, detected by the auxiliary LLM.
- Even when concepts are correlated, an explanation can mention them individually if they are truly influential.
- If the model used both Employment and Savings non-additively, a faithful explanation would mention *both*. The per-concept EE captures that without modification.
- **Faithfulness:** $F_{4\mathrm{CF}}(x) = \mathrm{PCC}(\mathrm{CE}_{4\mathrm{CF}}, \mathrm{EE})$, estimated via Bayesian hierarchical regression (same as original).

$\Rightarrow$ The entire pipeline remains identical except for how CE is computed for correlated groups.

4CF Meets All Four Requirements

1. **Handles correlations** – Shapley values capture synergy, redundancy and independence [1].
2. **Backwards compatible** – If no correlations exist, $\phi_m = \mathrm{CE}_{\mathsf{orig}}(C_m)$.
3. **Range** $[-1, 1]$ – PCC between two vectors, exactly as before.
4. **Feasible complexity** – Only compute Shapley for small groups ($|S| \leq 4$); total counterfactuals increase but stay manageable.

Computational Concept

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Claim: If concepts in a group S are independent (no interaction), then $\phi_m = \mathrm{CE}_{\text{orig}}(x, C_m) = v(\{m\})$.

Step 1 – Shapley definition:

$$\phi_m = \sum_{T \subseteq S \setminus \{m\}} \frac{|T|!\,(|S| - |T| - 1)!}{|S|!} \left[\, v(T \cup \{m\}) - v(T) \,\right]$$

Step 2 – Independence (additivity): For any disjoint $A, B \subseteq S$, $v(A \cup B) = v(A) + v(B)$. Therefore

$$v(T \cup \{m\}) - v(T) = v(\{m\}) \quad \forall\, T.$$

Step 3 – Substitute into the Shapley sum:

$$\phi_m = v(\{m\}) \cdot \underbrace{\sum_{T \subseteq S \setminus \{m\}} \frac{|T|!\,(|S| - |T| - 1)!}{|S|!}}_{=1}.$$

Step 4 – The weights sum to 1: The sum of the multinomial coefficients over all subsets equals 1 (it's a partition of all permutations of S).

$$\sum_{k=0}^{|S|-1} \binom{|S| - 1}{k} \frac{k!\,(|S| - k - 1)!}{|S|!} = 1.$$

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Hence $\phi_m = v(\{m\})$, which is exactly the original atomic CE. $\square$

Consequence: When the oracle finds no correlations, $\mathrm{CE}_{\mathrm{4CF}} = \mathrm{CE}_{\mathrm{orig}}$, so $F_{\mathrm{4CF}}(x) = F_{\mathrm{orig}}(x)$. Also if the oracle falsely "gives" a non-correlated concept to Shapley-calculation, the CE-value is not distorted.

How to Detect Correlated Concept Groups?

The "oracle" can be realised with several practical approaches:

1. **Pointwise Mutual Information (PMI):** compute co-occurrence of concept values across the dataset; high PMI $\Rightarrow$ group.
2. **Statistical tests:** ANOVA / χ^2 on concept value co-occurrences.
3. **Auxiliary LLM:** prompt an LLM to flag which concepts are likely correlated in meaning.

In the thesis, we will compare these methods experimentally. For this presentation, we assume a reliable oracle exists.

Experimental Setup

We re-evaluate faithfulness on three datasets:

- **BBQ (Social Bias)** – concepts mostly independent → scores should nearly match original CCF.
- **MedQA (Medical QA)** – some high-correlation questions (e.g. symptoms and diagnoses) → 4CF reveals differences where original underestimates.
- **German Credit** – tabular data with strong presumed correlations (e.g. 'Gastarbeiter' vs. 'Beruf') → large deviations expected; also eliminates need for many auxiliary LLM calls.

We also test robustness by increasing the number of questions.

Limitations of 4CF

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- **Computational cost:** Shapley value requires $2^{|S|}$ joint interventions per group. Limiting $|S| \leq 4$ keeps it feasible, but larger groups become expensive.
- **Open edges & transitivity:** Statistical correlation is not mathematically transitive. In practice however, for many social concepts (e.g., Nationality–Mother tongue–Ethnicity) transitivity can be assumed.
- **Oracle quality:** The accuracy of the correlation detector directly affects the group partitioning. A too reluctant oracle misses correlation and skews the result, a too aggressive oracle increases the computing complexity.
- **Factorial counterfactual generation:** Jointly swapping multiple concepts while preserving textual coherency of counterfactuals could introduce new challenges for the auxiliary LLM, but template-based approaches help for tabular data.

[1] Jan Ittner et al. *Feature Synergy, Redundancy, and Independence in Global Model Explanations Using SHAP Vector Decomposition*. Comment: 7 pages, 2 figures. July 26, 2021. DOI: 10.48550/arXiv.2107.12436. arXiv: 2107.12436 [cs]. URL: http://arxiv.org/abs/2107.12436 (visited on 05/12/2026). Pre-published.

[2] Katie Matton, Robert Ness, and John Guttag. "Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations". In: The Thirteenth International Conference on Learning Representations. Oct. 4, 2024. URL: https://openreview.net/forum?id=4ub9gpx9xw (visited on 04/24/2026).

[3] Christoph Molnar. *Interpretable Machine Learning (Third Edition)*. Dec. 20, 2025. URL: https://leanpub.com/interpretable-machine-learning (visited on 05/12/2026).

Thank you!

Questions?